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Power consumption (worst case):	
AMC3330QDWER	155 mW
AMC3301DWER	151 mW
AD5131E04IPAGR	10.2 mW
9-1462038-9 (Relay) x2	280 mW
1462039-4 (Relay) x2	200 mW
Total	796.2 mW
Budget:	5 W



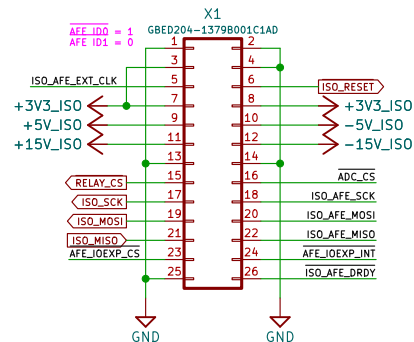
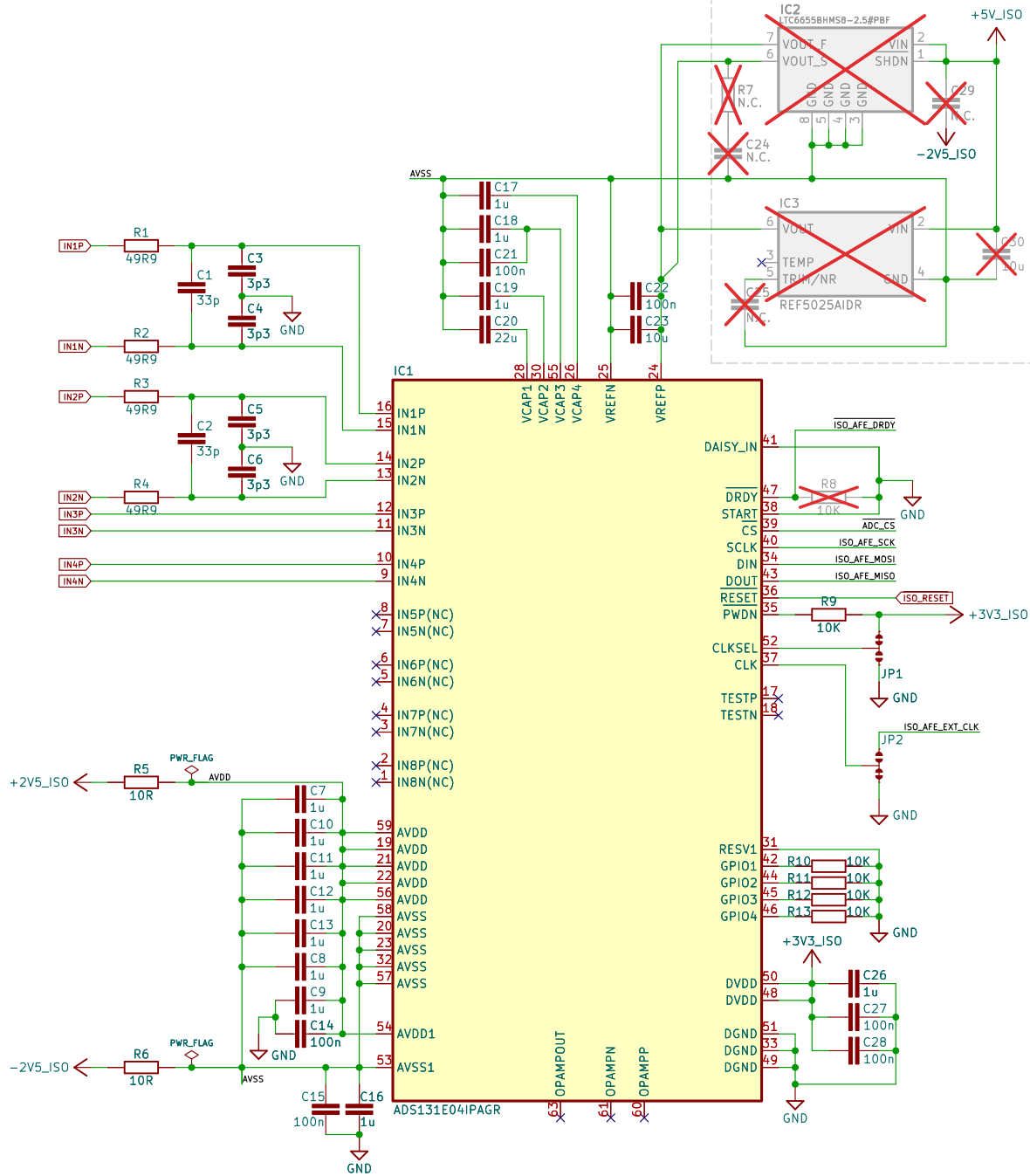
Id: 1/5

4-ch 24-bit ADC

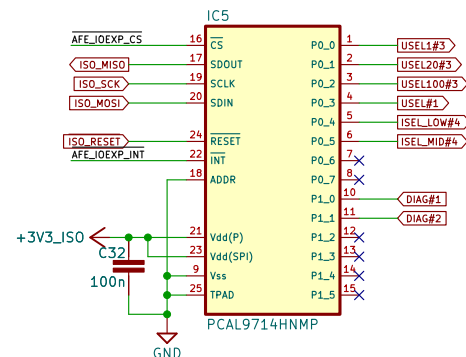
+2.5 V voltage reference (optional)

AFE connector

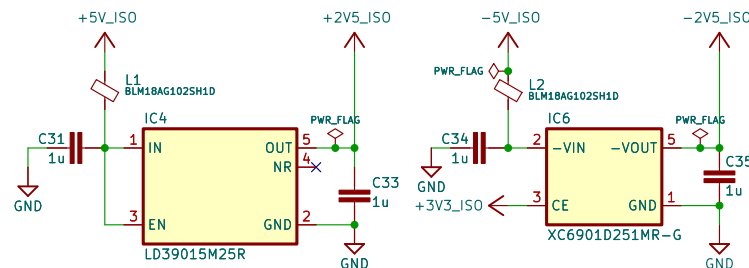
26-pin AFE connector



GND 1	2 GND
AFE_ID0 3	4 AFE_ID1
ISO_AFE_EXT_CLK 5	6 ISO_RESET
+3V3 7	8 +3V3
+5V 9	10 -5V
+15V 11	12 -15V
GND 13	14 GND
RELAY_CS 15	16 ADC_CS
SPI_SCK 17	18 AFE_CLK
SPI_MOSI 19	20 AFE_MOSI
SPI_MISO 21	22 AFE_MISO
AFE_IOEXP_CS 23	24 AFE_IOEXP_INT
GND 25	26 ISO_AFE_DRDY



+/-2.5 V LDOs



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4-channel Isolated AFE for MIO168+

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Sheet: /ADC, IOexp, LDOs/

File: ADC.kicad_sch

Title: EEZ DIB AFE3+

Size: A4

Date: 2026-08-23

KiCad E.D.A. 10.0.5

Rev: r1B1

Id: 2/5

Isolated Voltage and Current amplifier

8-pin isolated two channels connector

CH#2 CS- 1
CH#2 CS+ 2
CH#2 Vin- 3
CH#2 Vin+ 4
CH#1 CS- 5
CH#1 CS+ 6
CH#1 Vin- 7
CH#1 Vin+ 8

X2
8 AIN1P
7 AIN1N
6 AIN2P
5 AIN2N
4 AIN3P
3 IN3N
2 AIN4P
1 AIN4N

TBP02R1-381-08BE

F1 3561(KTR) x2
10A fuse

F2 3561(KTR) x2
1.25A fuse

K1B
K1C
K2C

R15 R200 1%
low TCR ($\pm 25\text{ppm}/^\circ\text{C}$)

R16 R022 1%
low TCR (max. $\pm 20\text{ppm}/^\circ\text{C}$)

L3
DLW215N3715Q2L

R19 10R
R20 10R

C38 8n2

L4
MLZ1608N100LT000

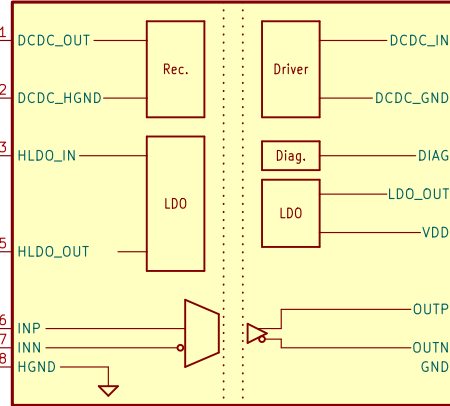
R21 1M5 0.1%
R22 1M5 0.1%
R23 1M5 0.1%
R24 82K5 0.1%

OK1 VOR1142B4

C39 1u
C43 1n
C44 100n
C41 100n
C47 1n

$\pm 1\text{ V max.}$

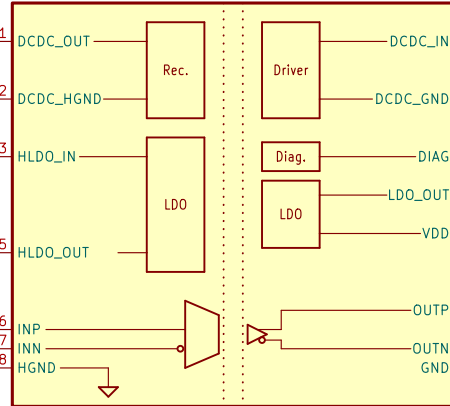
IC8



AMC3330QDWER

Fixed gain: x2
Voltage ranges: $\pm 50\text{ V}$, $\pm 480\text{ V}$

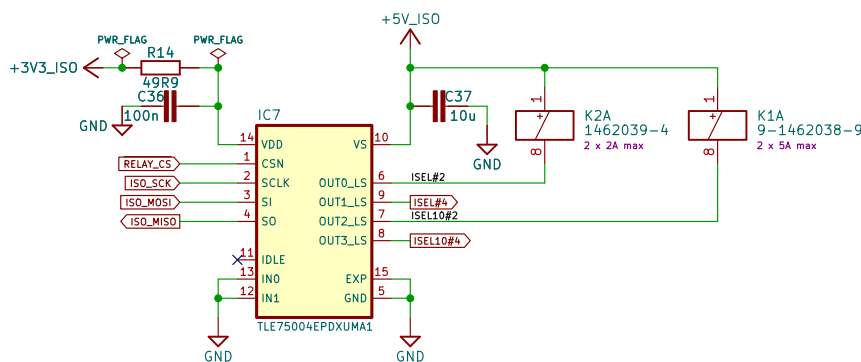
IC9



AMC3301DWER

Fixed gain: x8.2
Current ranges: $\pm 1\text{ A}$ (gain x1), $\pm 10\text{ A}$ (gain x2)

Relay driver



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4-channel Isolated AFE for MIO168+

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Sheet: /Isolated Ch#1, Ch#2/

File: Isolated_Ch#1,_Ch#2.kicad_sch

Title: EEZ DIB AFE3+

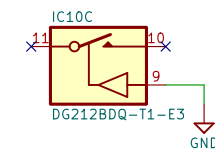
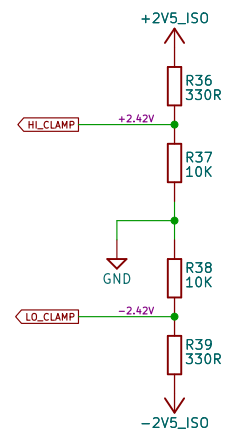
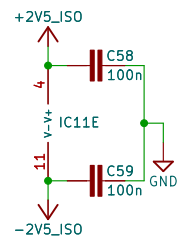
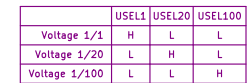
Size: A4

Date: 2026-08-23

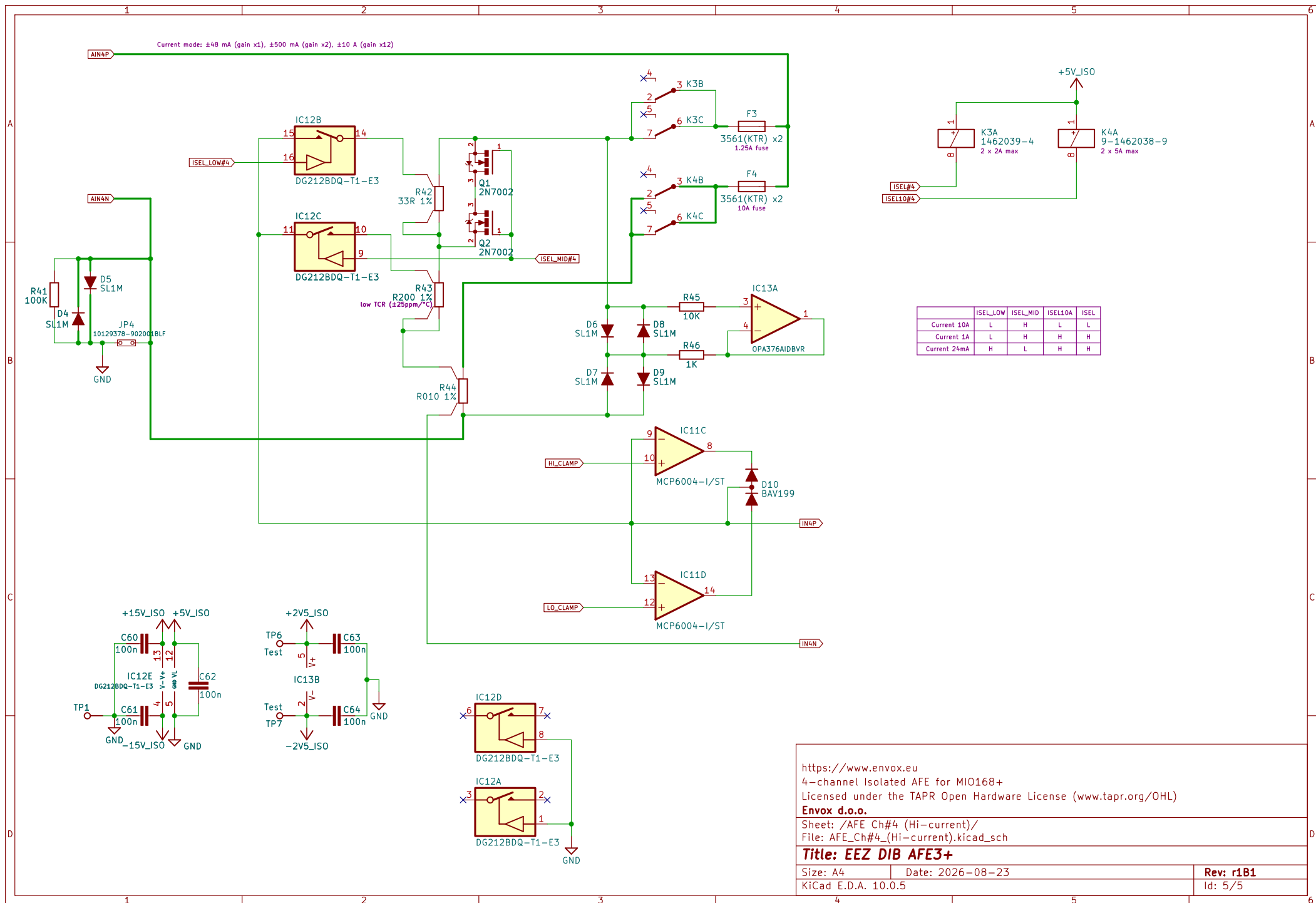
KiCad E.D.A. 10.0.5

Rev: r1B1

Id: 3/5



Rev: r1B1
Id: 4/5



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 4-channel Isolated AFE for MIO168+
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Sheet: /AFE Ch#4 (Hi-current)/
 File: AFE_Ch#4_(Hi-current).kicad_sch

Title: EEZ DIB AFE3+

Size: A4 Date: 2026-08-23
 KiCad E.D.A. 10.0.5

Rev: r1B1
 Id: 5/5